SYNASPOT: A LIGHTWEIGHT, STREAMING MULTI-MODAL FRAMEWORK FOR KEYWORD SPOTTING WITH AUDIO-TEXT SYNERGY

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ABSTRACT

Open-vocabulary keyword spotting (KWS) in continuous speech streams holds significant practical value across a wide range of real-world applications. While increasing attention has been paid to the role of different modalities in KWS, their effectiveness has been acknowledged. However, the increased parameter cost from multimodal integration and the constraints of end-to-end deployment have limited the practical applicability of such models. To address these challenges, we propose a lightweight, streaming multi-modal framework. First, we focus on multimodal enrollment features and reduce speaker-specific (voiceprint) information in the speech enrollment to extract speaker-irrelevant characteristics. Second, we effectively fuse speech and text features. Finally, we introduce a streaming decoding framework that only requires the encoder to extract features, which are then mathematically decoded with our three modal representations. Experiments on LibriPhase and WenetPrase demonstrate the performance of our model. Compared to existing streaming approaches, our method achieves better performance with significantly fewer parameters.

***Index Terms*—** Keyword spotting, open vocabulary, contrastive learning, streaming.

1. INTRODUCTION

Keyword spotting (KWS) is a technology designed to continuously detect the presence of keywords in audio streams, it is now serving as a critical entrance to human-machine interaction systems. Since KWS is typically deployed on resource-constrained devices, models must be compact and low-latency, which has driven the development of small-footprint, streaming KWS.

KWS systems broadly fall into two categories. The first is customized KWS (e.g., "Ok Google"), which is widely used for device wake-up. The other one is open-vocabulary KWS, it is emerged to improve user flexibility, allowing users to specify keywords on demand. Our work focuses on the latter. Existing open-vocabulary KWS approaches include audio enrollment methods such as Dynamic Time Warping (DTW) [1, 2] and metric learning [3], as well as text enrollment [4, 5, 6]. It has been shown that multi-modal enrollment features can increase both the expressiveness and accuracy of KWS models. In particular, cross-modal alignment features enhance performance overall, and treating audio alignment as auxiliary supervision offers further improvements [4]. Combining several aligned modalities delivers additional representational benefits [6]. Standing on these insights, we equip the proposed framework, SYNASPOT, with multi-modal enrollment (audio, text, and mixed), offering users a flexible choice of enrollment modes and enabling robust operation across diverse scenarios. For audio-only enrollment, we develop an audio encoder and enhance it with speaker domain adaptation. For text-only enrollment, we apply contrastive learning (CL) to align text feature space. When both modalities are available, we further incorporate a fused feature that combines audio and text modalities, increasing enrollment diversity and enhancing KWS accuracy.

To compare enrolled keywords with user-spoken speech, many existing decision methods rely on non-streaming end-to-end models, such as GRU combined with fully connected layers, for discrimination[7, 8, 9]. However, such models require audio segments with precisely aligned word boundaries, which makes them unsuitable for real-world streaming scenarios. However, many lightweight streaming models[10, 11] are limited to customized KWS, lacking open-vocabulary capability. To address this limitation, window-based streaming approaches have been proposed, where end-to-end decisions are performed over fixed-length windows[12]. Yet, this design leads to significant performance variations when the lengths of keywords differ. To overcome these issues, we propose a decoding framework that leverages only encoder representations and enrollment features for mathematical decoding. This enables the recognition of enrolled keyword with varying lengths directly from streaming audio, while maintaining a lightweight model that consists of a single encoder. Finally, we introduce a complete architecture that integrates multimodal enrollment features with streaming audio decoding. This design achieves a lightweight KWS system with both low latency and compact model size.

In summary, our contributions are concluded as follows:

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- We propose SYNASPOT, a lightweight, streaming multi-modal framework for KWS, fusing three modalities (audio, text, audio-text mixed).
- We develop a multimodal flexible enrollment keyword spotting system.
- We conduct series of experiments to validate the effectiveness of SYNASPOT. The results demonstrate that SYNASPOT outperforms baselines in both English and Chinese KWS tasks.

2. SYNASPOT: DESIGN DETAILS

As shown is Fig. 1, SYNASPOT consists of two phases, i.e., training phase, and inference & decoding phase.

2.1. Training Phase

2.1.1. Audio Embedding Modeling

In the training phase, we first learn a speaker-irrelevant audio encoder and explicitly enlarge the inter-phoneme margins.

The audio encoder comprises L DFSMN [13] layers. It takes audio FBank frames as input and outputs frame-level audio embeddings E_A. A phoneme classifier is then applied to E_A to predict the label of each frame. To reduce phoneme confusion and false positives, we adopt the additive angular margin (AAM) loss [14]. The objective is defined in Function 1, where θ_j represents the angle between the model weight vector $\mathbf{w}_j$ and the input x_i, with s serving as the rescaling parameter, and m acting as the angular margin penalty.

$$\mathcal{L}_{\text{ph}} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{e^{s(\cos(\theta_{y_i}+q))}}{e^{s(\cos(\theta_{y_i}+q))}+\sum_{j=1,j\neq y_i}^{l}e^{s(\cos(\theta_j))}} \tag{1}$$

To reduce speaker-specific (voiceprint) information, we apply domain adaptation [15] to SYNASPOT by attaching a speaker classifier to the audio encoder and reversing its gradients during training. This encourages utterances with the same linguistic content but different speakers to produce similar embeddings, thereby improving the effectiveness of enrollment audio in the keyword spotting task. The speaker classifier consists of an attentive pooling layer [14] followed by a linear classifier and is optimized using the cross-entropy loss, denoted as $\mathcal{L}_{vp}$.

We jointly optimize the two objectives as shown in Function 2, where α_A and β_A are hyper-parameters that control their relative contributions.

$$\mathcal{L}_{\text{audio}} = \alpha_A \cdot \mathcal{L}_{\text{ph}} + \beta_A \cdot \mathcal{L}_{\text{vp}} \tag{2}$$

2.1.2. Text & Mixed Embedding Modeling

We obtain text embeddings E_T using an embedding layer followed by an LSTM [16]. To form the mixed embedding E_M, we apply a cross-attention layer in which E_T serves as the queries and E_A serves as the keys and values. We then align the audio, text and mixed modalities in a shared embedding space via contrastive learning [5]. The objectives are defined in Function 3 and Function 4, where sim denotes the similarity function (cosine similarity on L2-normalized embeddings unless otherwise noted), E_*^i means the i^{th} frame of embedding E_*, τ>0 is the temperature, and $i \neq j$.

$$\mathcal{L}_{\text{clat}} = -\sum_{i=1}^{N}\log\frac{\exp(sim(E_T^i, E_A^i)/\tau)}{\sum_{j=1}^{N}\exp(sim(E_T^i, E_A^j)/\tau)} \tag{3}$$

$$\mathcal{L}_{\text{clam}} = -\sum_{i=1}^{N}\log\frac{\exp(sim(E_M^i, E_A^i)/\tau)}{\sum_{j=1}^{N}\exp(sim(E_M^i, E_A^j)/\tau)} \tag{4}$$

We use the audio encoder trained in Section 2.1.1 and jointly optimize the three objectives as shown in Function 5, where α_M, β_M and γ_M are hyper-parameters that control their relative contributions.

$$\mathcal{L}_{\text{mixed}} = \alpha_M \cdot \mathcal{L}_{\text{ph}} + \beta_M \cdot \mathcal{L}_{\text{clat}} + \gamma_M \cdot \mathcal{L}_{\text{clam}} \tag{5}$$

2.2. Inference & Decoding Phase

To enable streaming and lightweight keyword spotting, SYNASPOT first computes and caches three enrollment embeddings from the enrollment audio and text, i.e., the audio embedding E_A, the text embedding E_T, and the mixed audio–text embedding E_M. Once verified, these embeddings are fixed and need not be recomputed. Consequently, during online inference the only model executed is the audio encoder. SYNASPOT processes the incoming waveform in small, contiguous audio chunks, computes frame-level audio embeddings E_W on the fly, and forwards them to the decoder to produce per-frame keyword scores.

In the decoding phase, we compute framewise similarity scores P between the streaming audio embeddings E_W and each enrollment embedding $E_{\text{Enroll}} \in \{E_A, E_T, E_M\}$. Let p_{ij} denote the similarity between E_{Enroll}^i and E_W^j.

Since raw scores can be noisy, we apply causal smoothing [17] using a moving average:

$$p'_{ij} = \frac{1}{j - h_{\text{smooth}} + 1}\sum_{k=h_{\text{smooth}}}^{j} p_{ik} \tag{6}$$

where $h_{\text{smooth}} = max\{1, j - w_{\text{smooth}} + 1\}$ is the index of the first frame within the smoothing window of size w_{smooth}.

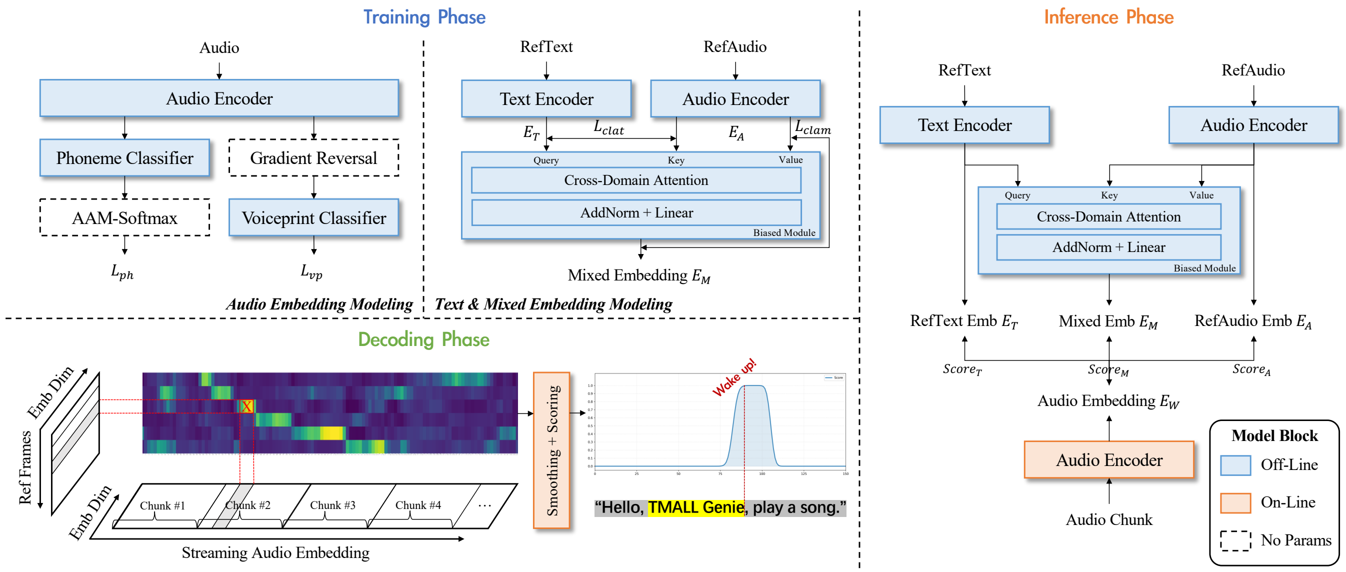

Fig. 1. An overview of SYNASPOT. In ① the training phase, we first learn a speaker-irrelevant audio encoder and explicitly enlarge the inter-phoneme margins, then we obtain text & mixed embeddings and align these modalities in a shared embedding space via contrastive learning. In ② the inference & decoding phase, we perform a streaming and lightweight keyword spotting, assign each audio frame with scores.

We then compute a confidence score at the j^{th} frame by aggregating the best per-unit similarities within a causal scoring window:

$$Score = \sqrt[n-1]{\prod_{i=1}^{n-1} \max_{h_{\text{scoring}} \leq k \leq j} p'_{ik}} \tag{7}$$

where $h_{\text{scoring}} = max\{1, j - w_{\text{scoring}} + 1\}$ is the index of the first frame within the scoring window of size w_{scoring}.

Depending on the enrollment modality, we obtain $Score_A$ (audio-only), $Score_T$ (text-only), or $Score_{All} = \alpha_S \cdot Score_A + \beta_S \cdot Score_T + \gamma_S \cdot Score_M$ (audio + text + mixed), where α_S, β_S and γ_S are fusion weights.

Table 1. Overall Effectiveness.

Model	EM†	Params	EER(%) LP$_E$‡	EER(%) LP$_H$‡	AUC(%) LP$_E$	AUC(%) LP$_H$
CMCD [9]	T	0.7M	8.42	32.90	96.70	73.58
Triplet [18]	T	0.6M	32.75	44.36	63.53	54.88
SoftTripe [19]	T	N/A	28.74	41.95	78.74	62.65
InfoNCE [5]	T	2.2M	8.99	32.51	96.85	74.87
CLAD [5]	T	2.2M	8.65	30.30	97.03	76.15
Synaspot-AA¶	A	0.9M	8.85	32.14	96.15	73.75
Synaspot-AT¶	T	0.9M	7.07	28.69	97.17	77.35
Synaspot	TA	0.9M	**5.77**	**27.29**	**97.34**	**79.15**

†: Enrollment modality, T represents text, A represents audio.
‡: LP$_E$, LP$_H$ represent Libriphase Easy and Hard, respectively.
¶: A represents audio enrollment only, T represents text enrollment only

3. EXPERIMENTAL

3.1. Datasets

The LibriSpeech [20] train-clean-100 and train-clean-360 subsets are used for training. In addition, we employ the LibriPhrase [9] dataset, which consists of phrases extracted from LibriSpeech. The evaluation dataset originates from the train-others-500 subset. Negative samples are divided into two categories: easy (**LP$_E$**) and hard (**LP$_H$**) based on their Levenshtein distance [21], where the hard set (**LP$_H$**) contains a large number of phonetically or orthographically similar words.

For Mandarin validation, we employed the WenetPhrase dataset[4] containing two specialized subsets: WenetPhrase-Easy (**WP$_E$**) and WenetPhrase-Hard (**WP$_H$**). To simulate real-world streaming scenarios, we constructed the test set for decoding evaluation by randomly concatenating speech segments as acoustic buffers preceding and following both **WP$_E$** and **WP$_H$** utterances.

3.2. Setup

We implement a prototype of SYNASPOT in PyTorch. The audio encoder uses $L = 7$ DFSMN layers, each has a hidden dimension of 256. We train for 50 epochs on eight NVIDIA RTX 3090 GPUs using Adam with an initial learning rate of 1e-3, an annealing factor of 0.3, and a batch size of 100.

During multimodal training, we weight the losses in

Table 2. Chinese KWS Effectiveness.

Model	EM†	Params	EER(%) WP$_E$‡	EER(%) WP$_H$‡	AUC(%) WP$_E$	AUC(%) WP$_H$
MM-KWS[4]	T	3.9M	4.24	26.23	99.19	79.24
MM-KWS-1.5s	T	3.9M	11.51	31.80	95.62	73.54
MM-KWS-2.0s	T	3.9M	18.38	39.15	89.67	66.45
Synaspot-AA¶	A	0.9M	18.46	36.68	89.58	68.19
Synaspot-AT¶	T	0.9M	19.76	39.95	87.25	64.70
Synaspot	TA	0.9M	14.56	34.50	92.87	70.35

†: Enrollment modality, T represents text, A represents audio.
‡: $\mathrm{WP_E}$, $\mathrm{WP_H}$ represent WenetPhrase Easy and Hard, respectively.
¶: A represents audio enrollment only, T represents text enrollment only

Table 3. Ablation Study.

Ablation	EER(%) LP$_E$‡	EER(%) LP$_H$‡	AUC(%) LP$_E$	AUC(%) LP$_H$
Synaspot	**5.77**	**27.29**	**97.34**	**79.15**
w/o Mixed Embedding	7.07	29.09	97.04	76.85
w/o Speaker Classifier	8.85	32.04	95.87	72.90

‡: $\mathrm{LP_E}$, $\mathrm{LP_H}$ represent Libriphase Easy and Hard, respectively.

Function(2) and (5) with $(\alpha_A, \beta_A) = (0.5, 0.5)$ while $(\alpha_M, \beta_M, \gamma_M) = (0.4, 0.3, 0.3)$. At inference time, score-level fusion uses $(\alpha_S, \beta_S, \gamma_S) = (0.5, 0.25, 0.25)$.

3.3. Overall Effectiveness

We present the effectiveness of our streaming decoding approach in Table 1, evaluated using the AUC and EER metrics. To demonstrate streaming performance, each audio sample in the LibriPhrase dataset was randomly padded with additional audio segments both before and after. The results demonstrate that our Synaspot model achieves superior efficiency-performance balance, attaining 27.71% EER and 78.70% AUC on $\mathrm{LP_H}$ along with 5.97% EER and 97.19% AUC on $\mathrm{LP_E}$ under streaming conditions, while maintaining a compact parameter footprint of 0.9M.

We validated the feasibility of our streaming algorithm on the Mandarin WenetPhrase dataset. As shown in Table. 2, our results in Mandarin demonstrate the generalizability of our method.

3.4. Qualitative Analysis

We visualize the similarity heatmaps in Fig. 2, which are derived from the similarity matrices of both offline embeddings (from audio, text, and mixed modalities) and online encoder outputs. These heatmaps clearly illustrate distinct highlighted regions between positive and negative examples, demonstrating the feasibility of our streaming decoding approach. As

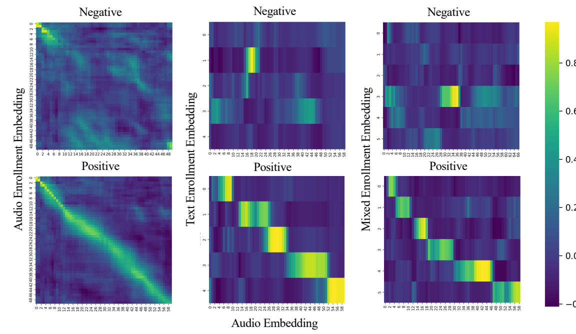

Fig. 2. Visualization of similarity heatmaps.

illustrated in Fig. 1. By leveraging the pre-registered horizontal coordinates of the three modalities for scoring, a final wake-up score is computed, resulting from the collaborative integration of all three modalities.

3.5. Ablation Study

Our ablation study results in Table 3 demonstrates that combining Mix Embedding and Speaker Classifiers improves Synaspot performance, confirming the effectiveness of the integrated feature representation. The architecture achieves speaker-discriminative separation while maintaining robustness to enrollment data variations.

3.6. stream to end-to-end

We present in Table 2 a systematic comparison between existing non-streaming end-to-end audio architectures and our proposed streaming framework. To highlight the limitations of its fixed-window approach, we evaluated the non-streaming model with a 2-second window, a 1.5-second window, and precise keyword boundaries. Our experiments demonstrate that traditional offline frameworks exhibit sensitivity to window length configurations and require processing 2-second audio segments, resulting in higher computational load and memory footprint. In contrast, streaming architectures enable frame-level audio input processing while achieving superior inference speed and enhanced operational flexibility through dynamic context adaptation.

4. CONCLUSION

In this paper, we introduce a lightweight and streaming multimodal framework for keyword spotting that effectively leverages audio-text synergy. The key innovation is the integration of multimodal features, speech, text, and fused representations, complemented by speaker separation for improved robustness. The resulting system is highly efficient and supports real-time streaming, making it ideal for on-device applications.

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